CAS SciFinder

Task History

Initiating Search

November 6, 2025, 4:37 PM

Search:

CAS Lexicon Search:

Hydrogenation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (126,585)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (2,059) Filtered By: Yield: **90-100%, 80-89%** Document Type: **Journal** Language: **English** Publication Year: **2020 to 2025**	Reactions	View Results

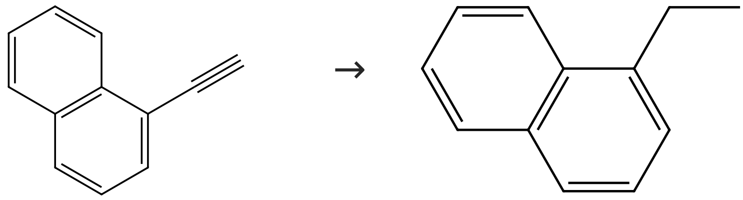
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (57)   Suppliers (62)

31-614-CAS-42142727   Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium, Carbon, Silica, Sulfuric acid, Polyaniline
Solvents: Methanol; 4 h, 30 bar, 50 °C

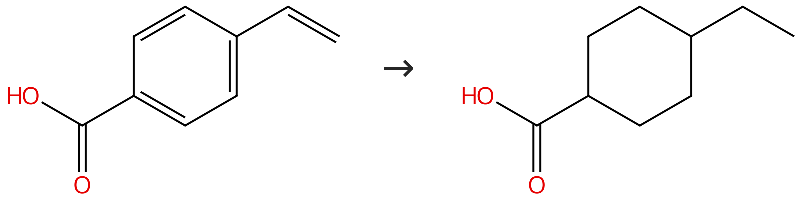
Catalytic Properties of Pd Deposited on Polyaniline in the Hydrogenation of Quinoline

By: Kompaniiets, Olena O.; et al

Chemistry (Basel, Switzerland) (2024), 6(4), 738-759.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (90)   Suppliers (57)

31-614-CAS-36090207   Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Platinum, Ruthenium
Solvents: Hexane; 24 h, 10 atm, 60 °C

Experimental Protocols

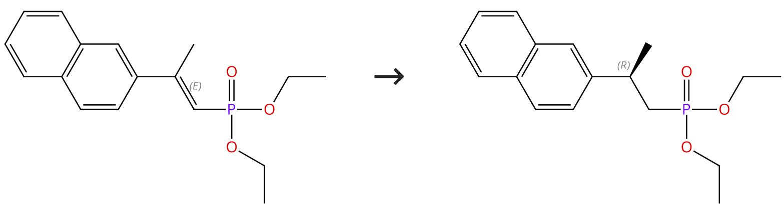
Intimate ruthenium-platinum nanoalloys supported on carbon catalyze the hydrogenation and one-pot hydrogenation-coupling reaction of oxidized amino derivatives

By: Rivero-Crespo, Miguel A.; et al

Catalysis Science & Technology (2023), 13(8), 2508-2516.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Double bond geometry shown

Absolute stereochemistry shown, Rotation (+)

31-242-CAS-20905772	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Rhodium(1+), bis[(1,2,5,6-η)-1,5-cyclooctadiene]-, tetrafluoroborate(1-) (1:1), (1*S*)-1-[(1*S*)-1-(Diphenylphosphino)ethyl]-2-[2-(diphenylphosphino)phenyl]ferrocene **Solvents:** Dichloromethane; 15 min, rt
1.2	**Reagents:** Hydrogen; 24 h, 10 bar, rt

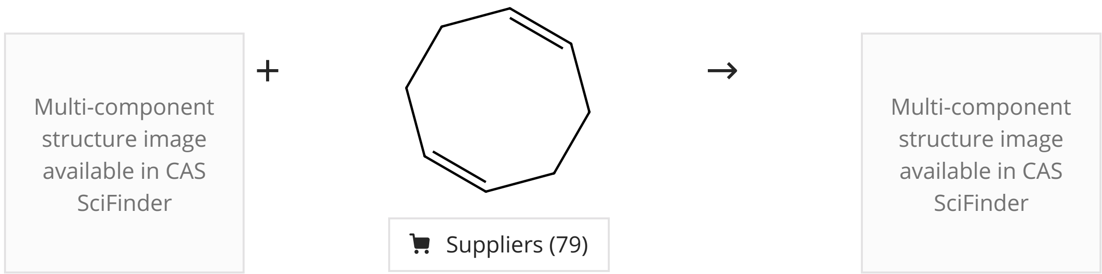
Experimental Protocols

Inverting External Asymmetric Induction via Selective Energy Transfer Catalysis: A Strategy to β-Chiral Phosphonate Antipodes

By: Onneken, Carina; et al

Angewandte Chemie, International Edition (2020), 59(1), 330-334.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



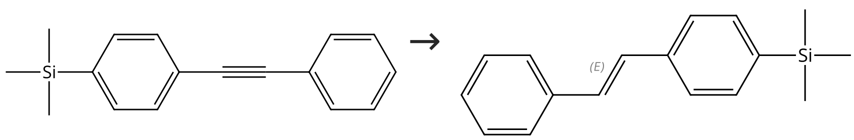
31-614-CAS-24186492	Steps: **1** Yield: **100%**
No Data Available	

η^2-Alkene Complexes of [Rh(PONOP-^{i}Pr)(L)]$^+$ Cations (L = COD, NBD, Ethene). Intramolecular Alkene-Assisted Hydrogenation and Dihydrogen Complex [Rh(PONOP-^{i}Pr)(η-H_2)]$^+$

By: Johnson, Alice; et al

Inorganic Chemistry (2021), 60(18), 13903-13912.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (52)

Double bond geometry shown

31-614-CAS-44553185	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** (*T*-4)-[2,6-Bis[[bis(1-methylethyl)phosphino-κ*P*]methyl]pyridine-κ*N*]methylcobalt **Solvents:** Benzene-d_6; 16 h, 23 °C

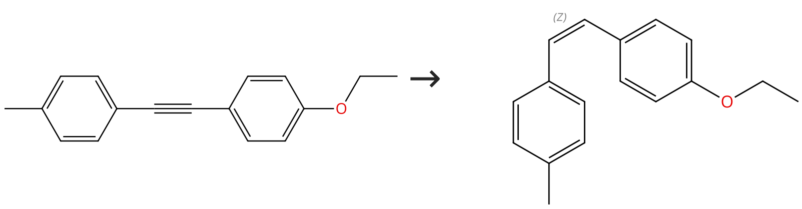
Experimental Protocols

Fast Co-Catalyzed Semihydrogenation of Alkynes with Controlled E/Z-Selectivity Using Same Catalyst

By: Pandey, Dilip K.; et al

ChemCatChem (2025), 17(8), e202500041.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (65)

Double bond geometry shown

31-614-CAS-41868086 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium *tert*-butoxide, Hydrogen
Catalysts: Tetrakis(acetonitrile)copper(1+) tetrafluoroborate
Solvents: Toluene; 16 h, 20 bar, rt

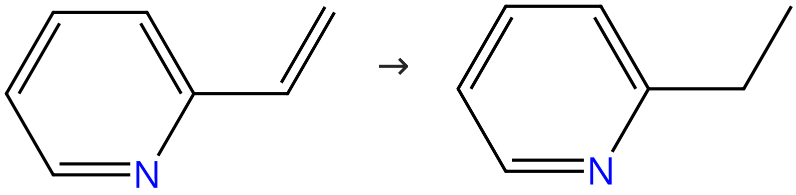
Experimental Protocols

(Semi)hydrogenation of enoates and alkynes effected by a versatile, particulate copper catalyst

By: Redl, Samuel; et al

Tetrahedron Chem (2024), 12, 100089.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (74) Suppliers (55)

31-614-CAS-38458080 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Platinum, 3018824-65-9
Solvents: Ethanol; 20 min, 1 bar, rt

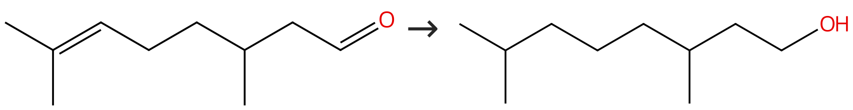
Experimental Protocols

Hydrogenation Catalysis by Hydrogen Spillover on Platinum-Functionalized Heterogeneous Boronic Acid-Polyoxometalates

By: Li, Shujun; et al

Angewandte Chemie, International Edition (2023), 62(50), e202314999.

Scheme 8 (2 Reactions) Steps: **1** Yield: **100%**



Suppliers (89) Suppliers (66)

31-242-CAS-23392837 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Ruthenium
Solvents: Ethyl acetate; 24 bar, 120 °C

Experimental Protocols

Continuous Flow Hydrogenation of Flavorings and Fragrances Using 3D-Printed Catalytic Static Mixers

By: Kundra, Milan; et al

Industrial & Engineering Chemistry Research (2021), 60(5), 1989-2002.

31-242-CAS-21444399 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Ruthenium
Solvents: Water; 270 min, 10 bar, 30 °C

Experimental Protocols

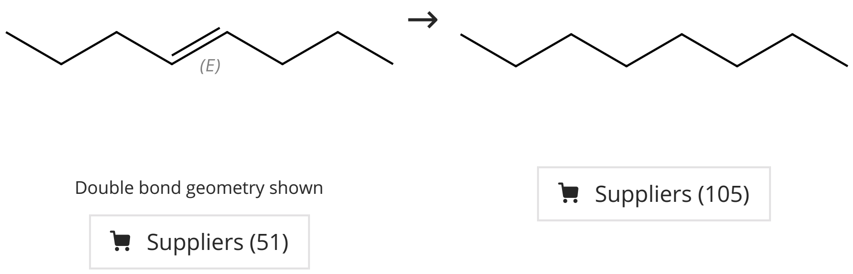
Anionic Amphiphilic Cyclodextrins Bearing Oleic Grafts for the Stabilization of Ruthenium Nanoparticles Efficient in Aqueous Catalytic Hydrogenation

By: Cocq, Aurelien; et al

ChemCatChem (2020), 12(4), 1013-1018.

Scheme 9 (1 Reaction)

Steps: **1** Yield: **100%**



31-614-CAS-37376089

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium; 3 h, 1 atm, 70 °C

Experimental Protocols

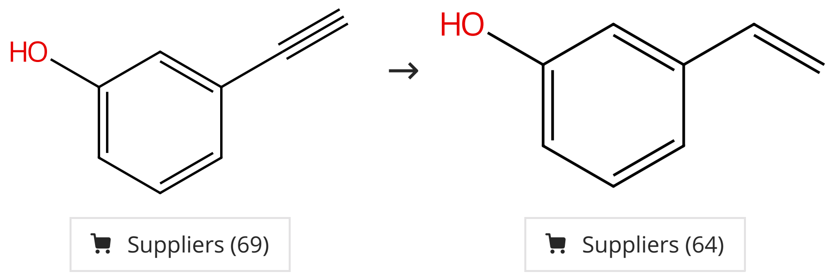
Co-generation of palladium nanoparticles and phosphate supported on metal-organic frameworks as hydrogenation catalysts

By: Takashima, Yohei; et al

Dalton Transactions (2023), 52(32), 11158-11162.

Scheme 10 (1 Reaction)

Steps: **1** Yield: **100%**



31-614-CAS-24553927

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium, Silver, Carbon
Solvents: Ethanol; 24 h, 1 atm, 25 °C

Experimental Protocols

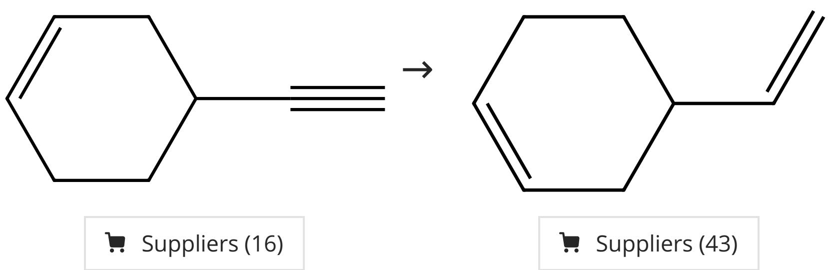
Seed-mediated Growth of Alloyed Ag-Pd Shells toward Alkyne Semi-hydrogenation Reactions under Mild Conditions

By: Zheng, Yuqin; et al

Chinese Journal of Chemistry (2021), 39(11), 3071-3078.

Scheme 11 (1 Reaction)

Steps: **1** Yield: **100%**



31-614-CAS-24553925

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium, Silver, Carbon
Solvents: Ethanol; 10 h, 1 atm, 25 °C

Experimental Protocols

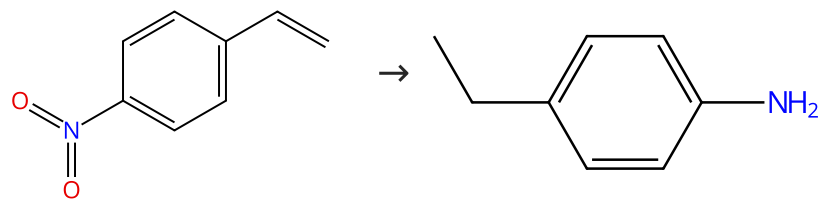
Seed-mediated Growth of Alloyed Ag-Pd Shells toward Alkyne Semi-hydrogenation Reactions under Mild Conditions

By: Zheng, Yuqin; et al

Chinese Journal of Chemistry (2021), 39(11), 3071-3078.

Scheme 12 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (64)

Suppliers (78)

31-614-CAS-43293765

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Cobalt oxide (CoO), Manganese oxide (Mn_3O_4), Cobalt
Solvents: Tetrahydrofuran, Water; 14 h, 5 bar, 100 °C

Experimental Protocols

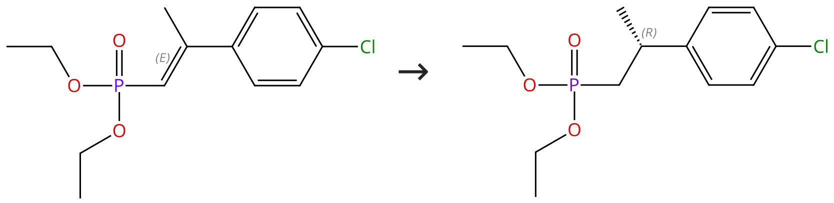
Synergistic Nanoscale Mn_3O_4-CoO-Co Heterojunctions for Boosting the Selectivity in Hydrogenation of Nitrostyrenes and Nitroarenes

By: Mrugesh, Padariya; et al

Chemistry - A European Journal (2024), 30(71), e202403236.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **100%**



Double bond geometry shown

Absolute stereochemistry shown, Rotation (+)

31-242-CAS-20905756

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Rhodium(1+), bis[(1,2,5,6-η)-1,5-cyclooctadiene]-, tetrafluoroborate(1-) (1:1), (1*S*)-1-[(1*S*)-1-(Diphenylphosphino)ethyl]-2-[2-(diphenylphosphino)phenyl]ferrocene
Solvents: Dichloromethane; 15 min, rt

1.2 **Reagents:** Hydrogen; 24 h, 10 bar, rt

Experimental Protocols

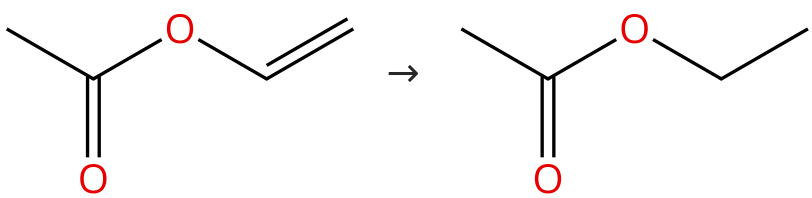
Inverting External Asymmetric Induction via Selective Energy Transfer Catalysis: A Strategy to β-Chiral Phosphonate Antipodes

By: Onneken, Carina; et al

Angewandte Chemie, International Edition (2020), 59(1), 330-334.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (69)

Suppliers (296)

31-242-CAS-23402969

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Nickel
Solvents: Ethanol; 20 bar, 120 °C

Experimental Protocols

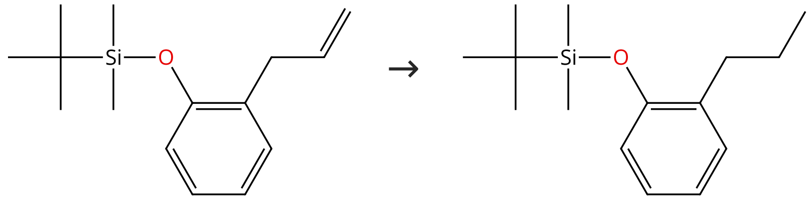
Continuous Flow Hydrogenation of Flavorings and Fragrances Using 3D-Printed Catalytic Static Mixers

By: Kundra, Milan; et al

Industrial & Engineering Chemistry Research (2021), 60(5), 1989-2002.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (3)

31-242-CAS-23727635

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen, 2641581-14-6
Catalysts: Palladium
Solvents: Water; 5 s, rt; 10 h, rt

Experimental Protocols

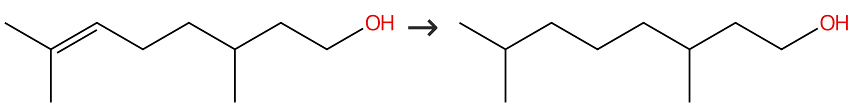
"TPG-lite": A new, simplified "designer" surfactant for general use in synthesis under micellar catalysis conditions in recyclable water

By: Thakore, Ruchita R.; et al

Tetrahedron (2021), 87, 132090.

Scheme 16 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (102)

Suppliers (66)

31-614-CAS-39874179

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Ruthenium, Silica
Solvents: Isopropanol; 3.5 h, 15 bar, 100 °C

Experimental Protocols

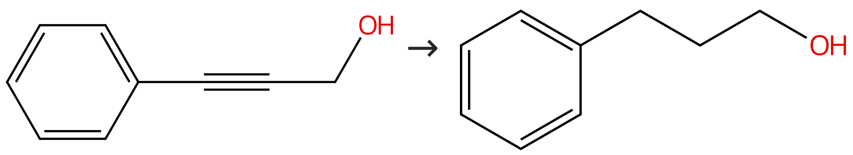
Ru Supported on Ni Phyllosilicate-Modified MCF as a Novel Nanocatalyst for Selective Hydrogenation of Biobased Citronellal to Citronellol

By: Mohire, Shalaka S.; et al

Industrial & Engineering Chemistry Research (2024), 63(6), 2619-2631.

Scheme 17 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (74)

Suppliers (87)

31-614-CAS-40885656

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Nickel, 3050811-37-2
Solvents: Water; 13 h, 10 bar, 35 °C

Experimental Protocols

Ni-NHC nanoparticles in micelles as an effective and reusable catalyst for hydrogenations and reductive-aminations in water

By: Avello, Marta G.; et al

ACS Sustainable Chemistry & Engineering (2024), 12(29), 10739-10751.

Scheme 18 (1 Reaction)

Steps: **1** Yield: **100%**

Suppliers (79)

Suppliers (43)

31-614-CAS-38458098

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Platinum, 3018824-65-9
Solvents: Ethanol; 60 min, 1 bar, rt

Experimental Protocols

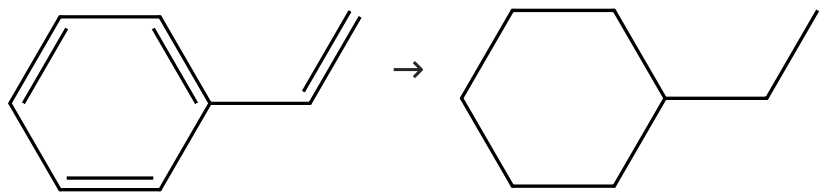
Hydrogenation Catalysis by Hydrogen Spillover on Platinum-Functionalized Heterogeneous Boronic Acid-Polyoxometalates

By: Li, Shujun; et al

Angewandte Chemie, International Edition (2023), 62(50), e202314999.

Scheme 19 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (115)

Suppliers (48)

31-242-CAS-21439030

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Ruthenium
Solvents: Water; 60 min, 10 bar, 30 °C

Experimental Protocols

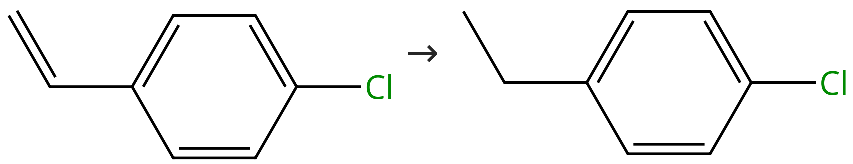
Anionic Amphiphilic Cyclodextrins Bearing Oleic Grafts for the Stabilization of Ruthenium Nanoparticles Efficient in Aqueous Catalytic Hydrogenation

By: Cocq, Aurelien; et al

ChemCatChem (2020), 12(4), 1013-1018.

Scheme 20 (2 Reactions)

Steps: **1** Yield: **100%**



Suppliers (78)

Suppliers (64)

31-614-CAS-38458072

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Platinum, 3018824-65-9
Solvents: Ethanol; 15 min, 1 bar, rt

Experimental Protocols

Hydrogenation Catalysis by Hydrogen Spillover on Platinum-Functionalized Heterogeneous Boronic Acid-Polyoxometalates

By: Li, Shujun; et al

Angewandte Chemie, International Edition (2023), 62(50), e202314999.

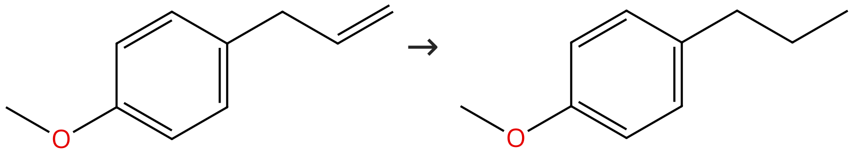
31-614-CAS-32040545	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Solvents:** Tetrahydrofuran; 10 h, 2 MPa, 80 °C
Experimental Protocols	

Solid surface frustrated Lewis pair constructed on layered Al OOH for hydrogenation reaction

By: Liu, Shulin; et al

Nature Communications (2022), 13(1), 2320.

Scheme 21 (2 Reactions) Steps: **1** Yield: **100%**



Suppliers (93) Suppliers (54)

31-614-CAS-38962305	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** 3026433-39-3 **Solvents:** Heptane, Water; 6 h, 900 psi, 90 °C
Experimental Protocols	

A combined catalytic and kinetic study of the aqueous-biphasic hydrogenation of allylbenzenes by using a dihydride-ruthenium complex containing tri(sodium-o-sulfonate dphenylphosphine) ligands

By: Baricelli, Pablo J.; et al

Journal of Organometallic Chemistry (2024), 1005, 122996.

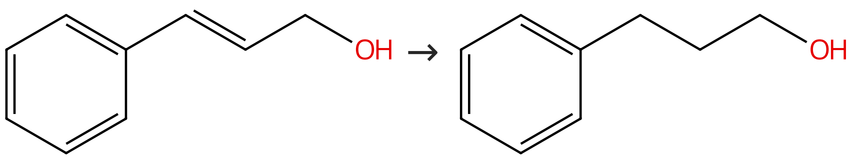
31-242-CAS-23389221	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Nickel **Solvents:** Ethyl acetate; 24 bar, 120 °C
Experimental Protocols	

Continuous Flow Hydrogenation of Flavorings and Fragrances Using 3D-Printed Catalytic Static Mixers

By: Kundra, Milan; et al

Industrial & Engineering Chemistry Research (2021), 60(5), 1989-2002.

Scheme 22 (2 Reactions) Steps: **1** Yield: **100%**



Suppliers (86) Suppliers (87)

31-614-CAS-34715630	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Palladium **Solvents:** Methanol; 4 h, 25 °C
Experimental Protocols	

Development of Silicon Carbide-Supported Palladium Catalysts and Their Application as Semihydrogenation Catalysts for Alkynes under Batch- and Continuous-Flow Conditions

By: Yamada, Tsuyoshi; et al

Catalysts (2022), 12(10), 1253.

31-242-CAS-21532799 Steps: **1** Yield: **100%**

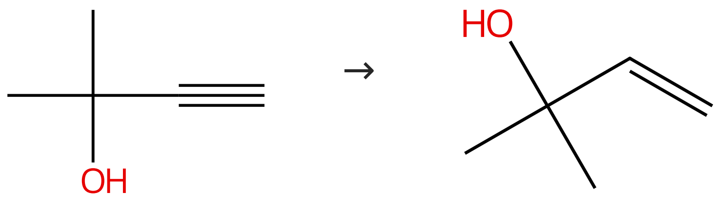
1.1 **Reagents:** Hydrogen
Solvents: Toluene; 8.8 min, 30 bar, 100 °C

Scrap waste automotive converters as efficient catalysts for the continuous-flow hydrogenations of biomass derived chemicals

By: Cova, Camilla Maria; et al

Green Chemistry (2020), 22(4), 1414-1423.

Scheme 23 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (66) Suppliers (69)

31-242-CAS-21535758 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium sulfide ($PdS_{0.53}$)
Solvents: Methanol, Dimethyl sulfoxide; 5 h, 1 atm, 50 °C

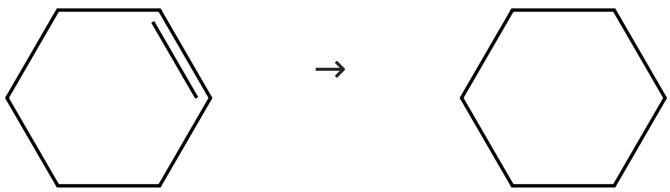
Experimental Protocols

Creation of Redox-Active PdS_x Nanoparticles Inside the Defect Pores of MOF UiO-66 with Unique Semihydrogenation Catalytic Properties

By: Dong, Ming-Jie; et al

Advanced Functional Materials (2020), 30(7), 1908519.

Scheme 24 (3 Reactions) Steps: **1** Yield: **100%**



Suppliers (63) Suppliers (214)

31-614-CAS-39167431 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Alumina
Solvents: Toluene; 8 h, 1 MPa, 100 °C

Experimental Protocols

Frustrated Lewis pairs on pentacoordinated Al^{3+}-enriched Al_2O_3 promote heterolytic hydrogen activation and hydroge nation

By: Wu, Qingyuan; et al

Chemical Science (2024), 15(9), 3140-3147.

31-614-CAS-38458066 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Platinum, 3018824-65-9
Solvents: Ethanol; 15 min, 1 bar, rt

Experimental Protocols

Hydrogenation Catalysis by Hydrogen Spillover on Platinum-Functionalized Heterogeneous Boronic Acid-Polyoxom etalates

By: Li, Shujun; et al

Angewandte Chemie, International Edition (2023), 62(50), e202314999.

31-614-CAS-34586721 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: 2839047-14-0
Solvents: Benzene-d_6; < 10 min, 1 atm, rt

Experimental Protocols

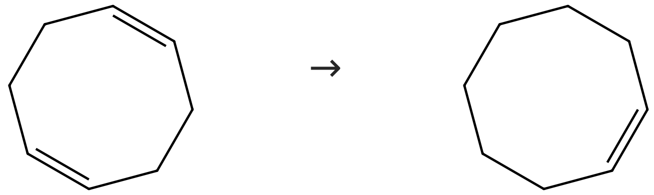
Iridium Complexes with a Naphthyridine-Based Si,N-Ligand: Synthesis and Catalytic Activity toward Olefin Hydrogenation

By: Sato, Keita; et al

Organometallics (2022), 41(18), 2612-2621.

Scheme 25 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (79) Suppliers (24)

31-614-CAS-24775226

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium
Solvents: Octane; 45 min, 50 °C

Experimental Protocols

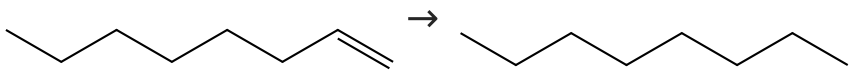
Synthesis of a hybrid Pd^0/Pd-carbide/carbon catalyst material with high selectivity for hydrogenation reactions

By: Garcia-Ortiz, Andrea; et al

Journal of Catalysis (2020), 389, 706-713.

Scheme 26 (5 Reactions)

Steps: **1** Yield: **100%**



Suppliers (93) Suppliers (105)

31-614-CAS-33720666

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: 1,2-Distearoyl-sn-*glycero*-3-phosphocholine, Palladium
Solvents: Water; 24 h, 1 atm, 22 °C

Experimental Protocols

Effects of lipid bilayer encapsulation and lipid composition on the catalytic activity and colloidal stability of hydrophobic palladium nanoparticles in water

By: Ortega, Dominick D.; et al

RSC Advances (2022), 12(34), 21866-21874.

31-242-CAS-24673760

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Rhodium
Solvents: Toluene; 5 h, 4.5 MPa, 80 °C

Experimental Protocols

Tandem hydroformylation/hydrogenation over novel immobilized Rh-containing catalysts based on tertiary amine-functionalized hybrid inorganic-organic materials

By: Gorbunov, Dmitry; et al

Applied Catalysis, A: General (2021), 623, 118266.

31-242-CAS-22784870

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Palladium
Solvents: Methanol; 80 min, 35 °C

Experimental Protocols

Palladium metallated shell layer of shell@core MOFs as an example of an efficient catalyst design strategy for effective olefin hydrogenation reaction

By: Gong, Yanyan; et al

Journal of Catalysis (2020), 392, 141-149.

31-242-CAS-21439026

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Ruthenium
Solvents: Water; 220 min, 10 bar, 30 °C

Experimental Protocols

Anionic Amphiphilic Cyclodextrins Bearing Oleic Grafts for the Stabilization of Ruthenium Nanoparticles Efficient in Aqueous Catalytic Hydrogenation

By: Cocq, Aurelien; et al

ChemCatChem (2020), 12(4), 1013-1018.

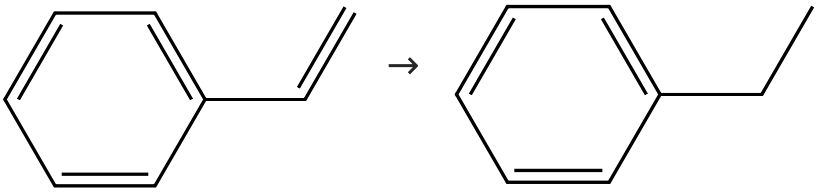
31-242-CAS-22390607	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Chloro(tetraphenylporphyrinato)rhodium (grafted on aminopropyltrimethoxysilane-modified SBA-15) **Solvents:** Isopropanol; 12 h, 10 bar, 100 °C
Experimental Protocols	

Rhodium porphyrin molecule-based catalysts for the hydrogenation of biomass derived levulinic acid to biofuel additive γ-valerolactone

By: Anjali, Kaiprathu; et al

New Journal of Chemistry (2020), 44(26), 11064-11075.

Scheme 27 (13 Reactions) Steps: **1** Yield: **100%**



Suppliers (115) Suppliers (133)

31-614-CAS-39640205	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Ruthenium, Aluminum nickel oxide (Al_2NiO_4) **Solvents:** Isopropanol, Water; 2 h, 7 bar, 200 °C
Experimental Protocols	

Water-assisted catalytic transfer hydrogenation of guaiacol to cyclohexanol over Ru/$NiAl_2O_4$

By: Zhao, Yuying; et al

Chemical Engineering Journal (Amsterdam, Netherlands) (2024), 485, 149934.

31-614-CAS-39551840	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Nickel(1+), [2-(dicyclohexylphosphino-κ*P*)-*N*-[2-(dicyclohexylphosphino-κ*P*)ethyl]ethanamine-κ*N*]hydro-, (*SP*-4-1)-, tetraphenylborate(1-) (1:1) **Solvents:** THF-d_8; 24 h, 4 atm, 80 °C

Mechanistic study of alkene hydrogenation catalyzed by cationic Ni(II)-hydride $PN^{H}P$ complexes - Ni(II/IV) vs. Ni(II) cycle

By: Wei, Zhihong; et al

Journal of Catalysis (2024), 431, 115372.

31-614-CAS-41868113	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium *tert*-butoxide, Hydrogen **Catalysts:** Tetrakis(acetonitrile)copper(1+) tetrafluoroborate **Solvents:** Toluene; 16 h, 20 bar, rt
Experimental Protocols	

(Semi)hydrogenation of enoates and alkynes effected by a versatile, particulate copper catalyst

By: Redl, Samuel; et al

Tetrahedron Chem (2024), 12, 100089.

31-614-CAS-42045448	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Palladium, Polydivinylbenzene; 2 h, 0.4 M Pa, 50 °C

Polymer-Supported Pd Nanoparticles for Solvent-Free Hydrogenation

By: Luo, Qingsong; et al

Journal of the American Chemical Society (2024), 146(38), 26379-26386.

31-614-CAS-38458070	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Platinum, 3018824-65-9 **Solvents:** Ethanol; 15 min, 1 bar, rt
Experimental Protocols	

Hydrogenation Catalysis by Hydrogen Spillover on Platinum-Functionalized Heterogeneous Boronic Acid-Polyoxometalates

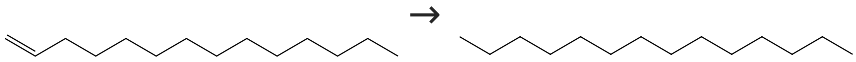
By: Li, Shujun; et al

Angewandte Chemie, International Edition (2023), 62(50), e202314999.

31-614-CAS-38458104 Steps: **1** Yield: **100%**	**Hydrogenation Catalysis by Hydrogen Spillover on Platinum-Functionalized Heterogeneous Boronic Acid-Polyoxometalates**
1.1 **Reagents:** Hydrogen **Catalysts:** Platinum, 3018824-63-7 **Solvents:** Ethanol; 15 min, 1 bar, rt Experimental Protocols	By: Li, Shujun; et al Angewandte Chemie, International Edition (2023), 62(50), e202314999.
31-614-CAS-36474585 Steps: **1** Yield: **100%**	**Air-Stable Efficient Nickel Catalyst for Hydrogenation of Organic Compounds**
1.1 **Reagents:** Hydrogen **Catalysts:** Nickel **Solvents:** Methanol; 24 h, 100 atm, 100 °C	By: Subotin, Vladyslav V.; et al Catalysts (2023), 13(4), 706.
31-614-CAS-32040542 Steps: **1** Yield: **100%**	**Solid surface frustrated Lewis pair constructed on layered AlOOH for hydrogenation reaction**
1.1 **Reagents:** Hydrogen **Solvents:** Tetrahydrofuran; 6 h, 2 MPa, 80 °C Experimental Protocols	By: Liu, Shulin; et al Nature Communications (2022), 13(1), 2320.
31-614-CAS-34586724 Steps: **1** Yield: **100%**	**Iridium Complexes with a Naphthyridine-Based Si,N-Ligand: Synthesis and Catalytic Activity toward Olefin Hydrogenation**
1.1 **Reagents:** Hydrogen **Catalysts:** 2839047-14-0 **Solvents:** Benzene-d_6; < 10 min, 1 atm, rt Experimental Protocols	By: Sato, Keita; et al Organometallics (2022), 41(18), 2612-2621.
31-614-CAS-31141641 Steps: **1** Yield: **100%**	**Rational design of Pt-Pd-Ni trimetallic nanocatalysts for room-temperature benzaldehyde and styrene hydrogenation**
1.1 **Reagents:** Hydrogen **Catalysts:** Nickel, Palladium, Platinum **Solvents:** Ethanol; 6.5 h, 1 MPa, 25 °C Experimental Protocols	By: Zheng, Tuo; et al Chemistry - An Asian Journal (2021), 16(16), 2298-2306.
31-614-CAS-25151593 Steps: **1** Yield: **100%**	**Embedded homogeneous ultra-fine Pd nanoparticles within MOF ultra-thin nanosheets for heterogeneous catalysis**
1.1 **Reagents:** Hydrogen **Catalysts:** Nickel acetate, Palladium chloride, Sodium borohydride, 5-(4-Pyridinylmethoxy)-1,3-benzenedicarboxylic acid **Solvents:** Ethanol; 50 min, 0.1 MPa, 25 °C Experimental Protocols	By: Guo, Taolian; et al Dalton Transactions (2021), 50(5), 1774-1779.
31-614-CAS-25139711 Steps: **1** Yield: **100%**	**Ligand-free, copper-free Sonogashira reaction and styrene hydrogenation catalyzed by 1-dodecanethiolate stabilized palladium nanoparticles**
1.1 **Reagents:** Hydrogen **Catalysts:** Palladium **Solvents:** Tetrahydrofuran; 2 h, rt Experimental Protocols	By: Lu, Feng Journal of Coordination Chemistry (2021), 74(15), 2534-2541.
31-242-CAS-22784871 Steps: **1** Yield: **100%**	**Palladium metallated shell layer of shell@core MOFs as an example of an efficient catalyst design strategy for effective olefin hydrogenation reaction**
1.1 **Reagents:** Hydrogen **Catalysts:** Palladium **Solvents:** Methanol; 80 min, 35 °C Experimental Protocols	By: Gong, Yanyan; et al Journal of Catalysis (2020), 392, 141-149.

Scheme 28 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (71)

Suppliers (106)

31-242-CAS-21439028

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Ruthenium
Solvents: Water; 3800 min, 10 bar, 30 °C

Experimental Protocols

Anionic Amphiphilic Cyclodextrins Bearing Oleic Grafts for the Stabilization of Ruthenium Nanoparticles Efficient in Aqueous Catalytic Hydrogenation

By: Cocq, Aurelien; et al

ChemCatChem (2020), 12(4), 1013-1018.